Foresight-TR6+: A Provenance-Audited, Leakage-Controlled Extension of the TriModal Ripeness 6 Dataset for Produce Spoilage Classification

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***Abstract*—Fruit and vegetable spoilage datasets are often reused without checking for duplicate images, conflicting labels or train–test leakage, which leads to inflated benchmark results. Foresight-TR6+ is a derived, metadata-only resource built on the public TriModal Ripeness 6 (TR-6) dataset (sRGB, thermal IR-fusion and methane recordings of banana, carrot, guava, Indian gooseberry, mango and tomato). A full audit of the 21,653 TR-6 files (19.77 GiB) found that every one of the 10,796 images in the *Classified* folder is a byte-identical copy of an image in the *Normal* folder, leaving 10,776 unique image contents, and that 170 thermal contents carry both *Spoiled* and *Not spoiled* labels. TR-6+ publishes a SHA-256 provenance manifest for every file, duplicate-group and label-conflict reports, a content-level manifest of 5,335 eligible sRGB images split 60/20/20 (3,177/1,087/1,071) by capture session with zero duplicate or session leakage, a stricter day-grouped sensitivity split, manifests for a reproducible 120-image noise-derivative pilot and 480 synthetic GAN images, an annotation protocol, and validation scripts. No source images are redistributed; users obtain them from the original TR-6 record. The resource enables reproducible, leakage-controlled benchmarking of image-based spoilage classifiers.**

***Index Terms*—Computer vision, Dataset descriptor, Data provenance, Duplicate detection, Food spoilage, Leakage-controlled split, Produce quality, TriModal Ripeness 6.**

I. Dataset Overview

Source dataset: https://doi.org/10.6084/m9.figshare.30783827.v1

Foresight-TR6+ is a derived dataset resource that makes the TriModal Ripeness 6 (TR-6) dataset [1] safe to use for supervised learning. TR-6 tracks six produce types from the unripe to the spoiled stage. It records 17 photographs per capture (16 side views at equal angles around the fruit and one top view) in three modalities: sRGB images, thermal IR-fusion images and methane gas-sensor logs.

TR-6+ does not add new photographs. It adds the information needed to use TR-6 correctly:

- an audited inventory with a SHA-256 hash for every file;
- explicit duplicate and label-conflict records;
- documented eligibility rules;
- reproducible train/validation/test partitions that keep near-identical views out of different partitions;
- manifests for derived (noisy) and synthetic (GAN) images;
- an annotation protocol and validation tooling.

The resource was produced as part of the Foresight project, a research prototype for visual produce-spoilage estimation.

II. Classes and Descriptions

Labels are organised along two dimensions: produce type and spoilage state.

A. Produce Types

- **Banana:** elongated fruit whose peel darkens from yellow to brown/black with progressive spotting.
- **Carrot:** root vegetable that develops dark surface lesions and shrivelling.
- **Guava:** green round fruit that develops brown spots and soft decayed patches.
- **Indian Gooseberry (Amla):** small, smooth green fruit; visible change during storage is subtle.
- **Mango:** large fruit that turns yellow-orange and develops dark blemishes.
- **Tomato:** red fruit that shows stem-scar decay and wrinkling.

B. Spoilage State

- **Not spoiled / Spoiled:** binary folder labels provided by the TR-6 authors under `Classified/`.
- For every produce type, the last *Not spoiled* capture precedes the first *Spoiled* capture. The label therefore marks a stage of the storage time series. It was **not** independently re-assessed by TR-6+.
- Images found only in `Normal/` carry no spoilage label and are marked `unlabeled`.
- Contents filed under both labels are marked `inherited_conflicting` and are never assigned a label automatically.

III. Data Format

The resource is a single ZIP archive (`Foresight-TR6-plus-dataset-v1.1.0.zip`, 88 files, 6.9 MB) of CSV, JSON, Markdown and Python files:

- **Master provenance manifest** (`manifests/master_manifest.csv`): one row per TR-6 file (21,653 rows), with these fields:

  - `record_id` and `content_id`;
  - `original_relative_path`;
  - `sha256` and `size_bytes`;
  - `produce_type` and `apparent_modality`;
  - `spoilage_label` and `label_status`;
  - `filename_timestamp`, `width`, `height`;
  - `duplicate_group_id` and `cross_label_conflict`;
  - the nullable `specimen_id` and `cross_modal_group_id`, left empty because TR-6 provides no such identifiers.
- **Duplicate and conflict reports** (`manifests/`): duplicate groups and their members, the label-conflict report, and modality and produce summaries.
- **ML split manifest** (`ml_splits/session_grouped_v1/split_manifest.csv`): one row per unique image content (10,776 rows), with these fields:
  - `sample_id`, `sha256` and `content_group_id`;
  - `source_relative_path` and `all_source_paths`;
  - `produce_class`, `source_label` and `label_status`;
  - `modality` and `capture_timestamp`;
  - `session_id` and `split` (`train`, `val`, `test` or `excluded`);
  - `exclusion_reason`, `annotation_status` and `seed`.

  A day-grouped sensitivity split uses the same schema.
- **Derived and synthetic manifests:**
  - `derived/noise-pilot-v1/` records the parent path and hash, derived hash, transformation, parameters, seed, inherited split and `is_synthetic=true` for each derivative.
  - `synthetic/gan/` records the provenance `synthetic_cdcgan_train_only` and quality checks.
- **Annotations** (`annotations/`): a Label Studio/COCO protocol, a status ledger and proposal-only polygons, which are explicitly not ground truth.
- **Scripts and integrity:** audit, split, validation, noise-generation and packaging scripts; `SHA256SUMS.txt` for every packaged file.

IV. Dataset Composition

Table I gives the leakage-controlled partition of the eligible sRGB contents. Table II gives the class distribution, and Table III summarises the audited source and the TR-6+ records.

V. Collection and Construction Methodology

The source images and sensor logs were collected by the TR-6 authors [1]: each fruit was imaged three times a day through natural decomposition. TR-6+ was constructed from a local, read-only copy of the published archive in six steps.

TABLE I
TR-6+ Session-Grouped Split of Eligible sRGB Images

Partition	Images	Sessions	Not spoiled	Spoiled	Share
Training	3,177	189	2,192	985	59.55%
Validation	1,087	64	747	340	20.37%
Testing	1,071	64	731	340	20.07%
Total	5,335	317	3,670	1,665	100%

TABLE II
Eligible sRGB Images per Produce Type and Spoilage State

Produce	Not spoiled	Spoiled	Total
Banana	153	170	323
Carrot	102	356	458
Guava	202	306	508
Indian Gooseberry	884	527	1,411
Mango	256	136	392
Tomato	2,073	170	2,243
Total	3,670	1,665	5,335

TABLE III
Audited Source and TR-6+ Record Specifications

Parameter	Value	Description
TR-6 files audited	21,653	19.77 GiB; 0 unreadable
sRGB images	10,676	4000×3000 JPEG (EXIF orientation 6)
Thermal IR-fusion images	10,950	640×480 palette JPEG
Methane logs	18 files	26,928 parsed records
Unique image contents	10,776	by SHA-256
Exact-duplicate groups	10,589	10,850 excess copies
Cross-label conflicts	170	all thermal (153 banana, 17 gooseberry)
Eligible sRGB contents	5,335	conflict-free, labelled
Excluded contents	5,441	5,084 thermal; 187 unlabelled; 170 conflicting
Capture sessions	317	$\leq$300 s gap, same date
Noise derivatives	120	24 sources $\times$ 5 transforms (manifest only)
Synthetic GAN images	480	64×64, train split only (manifest only)
Human-reviewed masks	0	12 machine proposals
Verified cross-modal pairs	0	not created without provenance

A. Construction Pipeline

1) **Audit:** every file was hashed with SHA-256 and decoded, and its modality, label, resolution and filename timestamp were recorded. An independent re-run of the audit reproduced all summary counts exactly.
2) **De-duplication:** byte-identical files were collapsed into one content record, so the `Classified` and `Normal` copies of an image can never be split apart.
3) **Eligibility:**
   - contents labelled both *Spoiled* and *Not spoiled* were excluded rather than tie-broken;
   - `Normal`-only contents were treated as unlabelled;
   - the ML manifest is restricted to sRGB, the modality

available from a phone camera.

4) **Grouping and split:**
   - images of the same produce and label whose timestamps are at most 300 s apart on the same date form a capture session, usually the 17 views of one fruit;
   - within each produce × label stratum, sessions were shuffled (seed 42) and assigned 60/20/20;
   - the day-grouped variant instead keeps whole capture days together.
5) **Derivatives:**
   - a deterministic noise pilot (Gaussian $\sigma = 8, 20$; salt-and-pepper 1%, 4%; speckle variance 0.04; seed 20261015) was generated for 24 pilot sources;
   - every derivative inherits its source's split and is flagged as not usable as an independent test specimen.
6) **Packaging:** absolute paths were removed, every file was checksummed, and the release was scanned to confirm it contains no image files.

B. Quality Assurance Workflow

1) **Integrity validation:** the validator found 21,653/21,653 source paths, recomputed 96/96 pilot hashes, and reported 0 errors.
2) **Leakage checks:** no content hash and no capture session occurs in more than one partition. In the day-grouped split only 1.5% of test images share a produce and capture day with training, against 76.2% in the session split.
3) **Reproducibility:**
   - re-running the split script yields a byte-identical manifest;
   - regenerating the noise pilot reproduced all 120 SHA-256 hashes.
4) **Annotation status:** machine-generated masks are labelled `proposal_only`; no annotation is described as ground truth unless a human has reviewed it.

VI. Potential Applications

1) **Leakage-controlled spoilage benchmarks:** training and comparing image classifiers for produce type and visible spoilage on fixed, documented partitions.
2) **Generalisation studies:** measuring how performance changes from session-grouped to day-grouped splits, i.e. generalisation to unseen storage days.
3) **Robustness and denoising research:** controlled noise derivatives with recorded parameters and seeds for robustness and denoising experiments.
4) **Synthetic-data evaluation:** studying whether GAN augmentation helps when synthetic images are kept strictly within the training partition.
5) **Dataset-auditing methodology:** a worked example of duplicate, conflict and provenance auditing for public multimodal datasets.

VII. Usage Notes

Obtaining images: download `TR-6.zip` from the source DOI. The manifests then resolve every record by its relative path and verify it by SHA-256.

Partition integrity: keep all images of a capture session (and preferably of a capture day) within one partition. Never treat `Classified` and `Normal` as independent samples.

Augmentation: apply augmentation after splitting and to training images only. Avoid hue or saturation shifts that alter colour-based decay cues.

Derived data: noisy and GAN images are synthetic and must not be used as validation or test specimens.

Reference results (Foresight project):

- A ViT-B/16 fine-tuned on the session-grouped split reached a test spoilage macro-F1 of 0.954 and a produce macro-F1 of 1.000.
- On the day-grouped split, the same ViT-S/16 configuration scored 0.098 lower than on the session-grouped split (mean of three seeds).

VIII. Limitations

Specimen identity: TR-6 provides no specimen identifiers. Each produce type is a small number of fruits photographed over consecutive days, so the same physical fruit appears in several partitions.

Label semantics: the spoilage labels are dataset-provided, date-ordered folder labels. Some *Spoiled* Indian gooseberries show no visible decay.

Domain: all images show a single fruit on a plain background under lab lighting, and class balance is skewed (tomato *Not spoiled* is 38.9% of eligible images).

Annotations: no human-reviewed segmentation masks exist yet, and no verified RGB–thermal–methane pairs exist.

No redistributed images: images are not included in the archive and must be obtained from the original source.

Licence and Citation

TR-6+ is released under CC BY 4.0, consistent with the licence of the source dataset, and requires attribution of the TR-6 authors [1]. An IEEE DataPort DOI will be added after publication.

References

[1] T. Bagyammal, Y. A. P. T. S, K. Deepak, and D. Unnikrishnan, "TriModal Ripeness 6," figshare, Dataset, 2025. doi: 10.6084/m9.figshare.30783827.v1.